

Intro to Cloud Computing

(CS4037) BDS8A

Date: May 27th, 2025

Course Instructor(s): Ali Khawaja

Final Exxam

Total Time (Hrs.): 3

Total Marks: 100

Total Questions: 5

Part 1 – Case Studies (60 Points total - 20 points each case study)

CLO 1. Apply cloud computing knowledge in various real world use cases (situations)

Q1. You are a developer in your organization and are involved in creating containerized solutions. You are tasked with creating an image of your application. Given the following requirements, write a complete Dockerfile for your container image:

- Your application is developed using dotnet framework. The webapp is compiled into webapp01.dll
- Your image uses ubuntu's base image with dotnet runtime installed: ubuntu/dotnet-runtime
- nginx web server is installed in your image. The web server is then configured with your application.
- Nginx will, by default run on port 5050, but the container will accept requests on port 80.
- Finally, configure your application to start whenever container is started.

Solution

- FROM ubuntu/dotnet-runtime
- apt install -y wget nginx
- WORKDIR /app
- COPY ./website/. .
- EXPOSE 80:5050
- ENTRYPOINT ["dotnet", "webapp01.dll"]

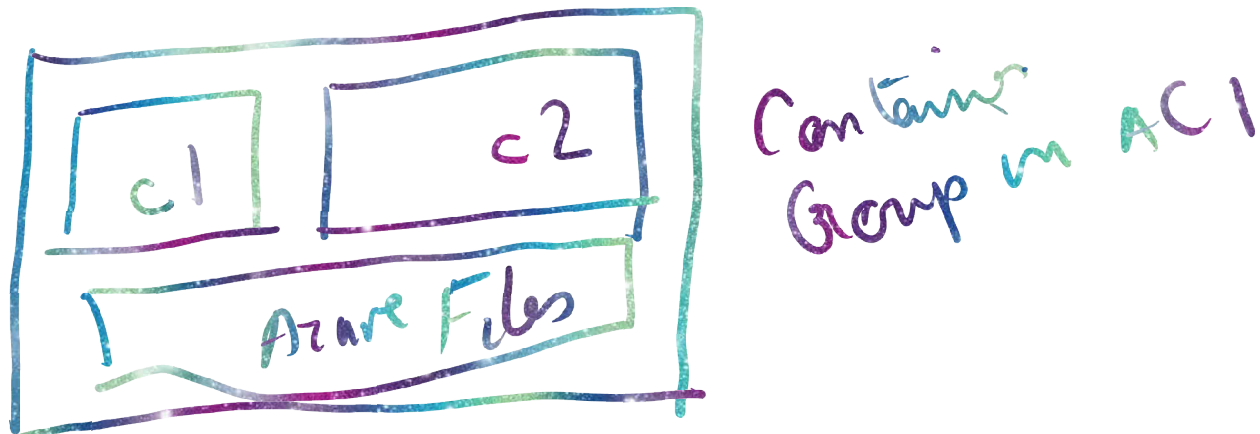
Steps 1,2,5,6: 4 Points each

Steps 3,4: 2 Points each

CLO 1. Apply cloud computing knowledge in various real world use cases (situations)

Q2. Your organization wants to deploy a simple containerized app to the cloud. The app is composed to two containers which must run together. Furthermore, the app must be publicly available. While the application does not handle much traffic, its critical enough that your organization wants to ensure that its up and running all the time, as well as restarted if it ever stops due to any failure. Both containers must persist all data in the storage to ensure no data gets lost in case of any container restart.

- Draw a detailed design diagram (7)
- Briefly list all design decisions to address all of the above requirements. (10)
- What are the deployment methodologies available and which is the preferred one? (3)



We'll use Azure Container Instances because:

1. Scalability not required.
2. Container Group can run both containers together
3. We can start restart policy on failure
4. Azure Files will be used to persist data from both containers

Deployment methodologies include ARM template and YAML file. YAML is the preferred format due to its concise nature to deploy containers.

CLO 1. Apply cloud computing knowledge in various real world use cases (situations)

Q3. FAST University is working on its next generation student registration system. The plan is to develop the front end using latest web development technologies while all the back end services will be developed as web services to provide various different functionalities.

The university wants to ensure it can manage, update, and scale each component independently and without affecting any other application area. Application has the following components:

1. Studentreg web application
2. erpsvc - backend app to integrate with backend applications and databases
3. nadarsvc – backend app to integrated with various government services
4. SQL Server – backend database (this will be configured on an IaaS VM)

As mentioned above, each part of the application must be manageable, updateable and can be scaled independently of all other parts. You also want to run multiple versions of the components if required. If multiple versions of web app are deployed, you should provide the ability for the users to test the newer version of the application.

Provide a clear design diagram of your solution. Clearly mention the options available and Address all the points mentioned in the problem. Clearly mention the features you will utilize to address various requirements.

Solution:

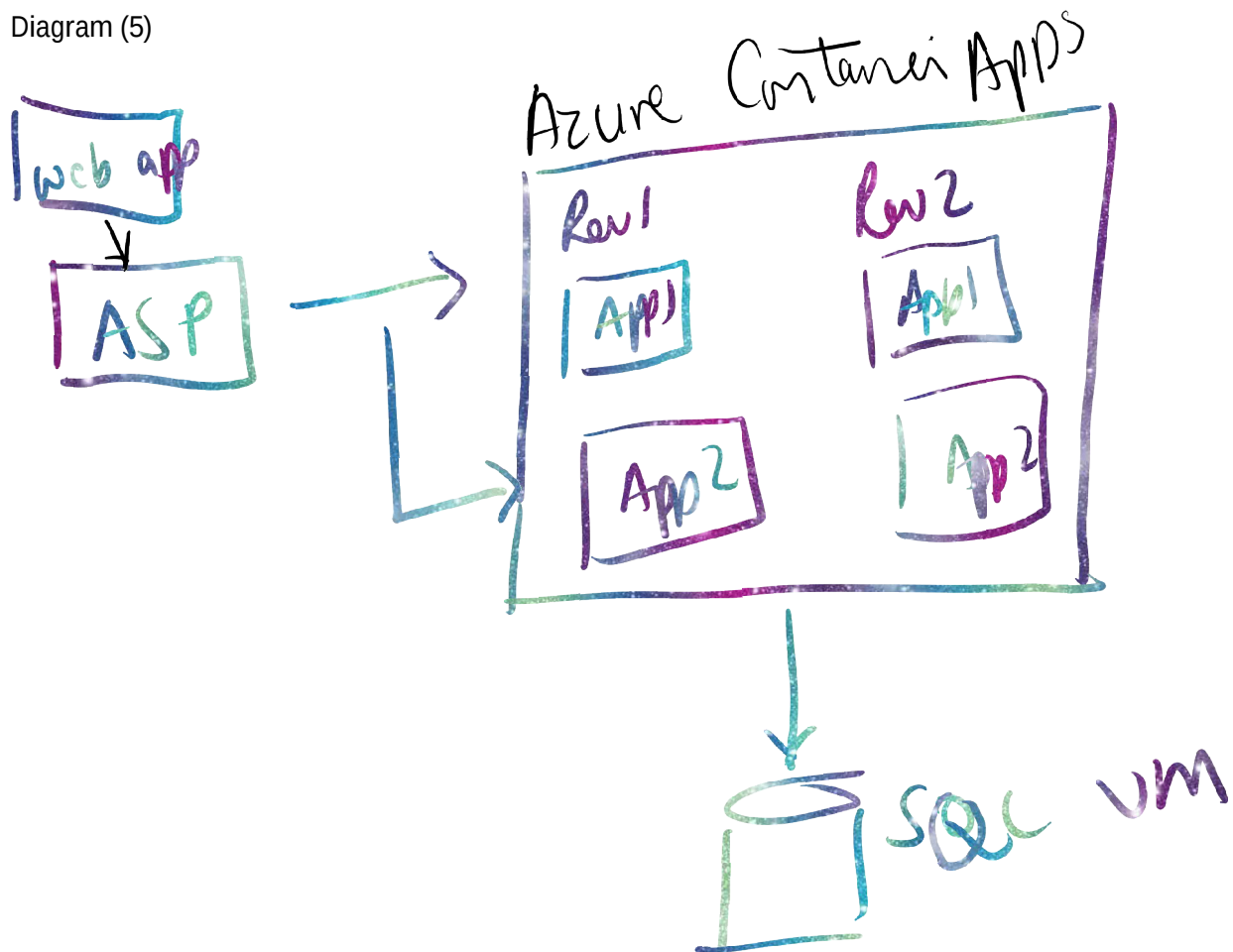
Design Decisions (15)

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1. 3 App components must be independently managed and scaled
2. We'll use Azure Container Apps for the two backend services as containers apps are well suits for back-end services due to its serverless nature
 - a. Containers apps will allow us to deploy multiple revisions for each
3. We'll use azure web apps for the front-end app
 - a. We'll use slots to deploy multiple versions
 - b. Uses can opt for the newer version. If they do, a cookie will be set in the http header for the application to route them to the other slot.
4. SQL Server will be deployed as a VM

Diagram (5)



Part 2 – Cloud Services (40 Points total - 20 points each)

Q4 – Azure Container Apps (2 Points Each, Option e,f: 4 Points each)

Provide brief and concise answers to all of the questions below

- a. Briefly and concisely explain Azure Container Apps PaaS service.

Ans: Azure Container Apps provides the flexibility you need with a serverless container service built for microservice applications and robust autoscaling capabilities without the overhead of managing complex infrastructure.

- b. It's an Azure Managed service over which platform?

Ans: Kubernetes

- c. Which open-source project is natively integrated with this platform to allow publish subscribe between the services, service to service invocation, etc?

Ans: DAPR

- d. Provide the cli commands to create and update a container app environment (Only the basic command without any options. Extra points will be awarded if options are provided)

Ans: az containerapp env create, az containerapp env update

- e. How do you verify that app was created successfully?

Ans: Select the link returned by the az containerapp create command to verify the container app is running.

- f. How can you deploy images from private registries in Azure Container apps?

Ans: You can deploy images hosted on private registries by providing credentials in the Container Apps configuration. To use a container registry, you define the required fields in registries array in the properties.configuration section of the container app resource template.

- g. What is a revision? What revision is used for? Briefly explain the purpose of revision-scope changes.

Ans: A revision is an immutable snapshot of a container app version. You can use revisions to release a new version of your app, or quickly revert to an earlier version of your app. New revisions are created when you update your application with [revision-scope changes](#).

- h. Which parameter in the azure container apps cli command allow you to specify confidential data such as passwords?

Ans: When you create a container app, secrets are defined using the `--secrets` parameter.

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Q5. Azure App Services (2 Points Each, Option e: 4 points)

Provide brief and concise answers to all the questions below. All questions are independent of each other.

- a. Azure app service plans provide the required features and compute for the deployed app services. When do you want to isolate your app in a new app service plan independently of other apps?

Ans: Isolate your app into a new App Service plan when:

- The app is resource-intensive.
- You want to scale the app independently from the other apps in the existing plan.
- The app needs resource in a different geographical region.

This way you can allocate a new set of resources for your app and gain greater control of your apps.

- b. If you have deployed 5 apps in an app service plan, and the app service plan is scaled to 3 instances, how those apps are distributed among the 3 VMs?

Ans: All web apps are deployed on all instances of the app service plan.

- c. How does automated deployment directly from Github works with Azure Web Apps?

Ans: When you connect your GitHub repository to Azure for automated deployment, any changes you push to your production branch on GitHub are automatically deployed for you.

- d. How do you deploy your app from command line in Azure App Service? What happens if the application does not exist?

Ans: webapp up is a feature of the az command-line interface that packages your app and deploys it. Unlike other deployment methods, az webapp up can create a new App Service web app for you.

- e. What are sidecar containers? And for what functionalities they are deployed for?

Ans: Sidecar containers let you deploy extra services and features to your container application without making them tightly coupled to your main application container. For example, you can add monitoring, logging, configuration, and networking services as sidecar containers.

- f. What is the purpose of federated identity in Azure App Service?

Ans: App Service uses federated identity, in which a third-party identity provider manages the user identities and authentication flow for you

- g. Regarding application authorization, what is the purpose of enabling “require authentication” in Azure App Service?

Ans: This option rejects any unauthenticated traffic to your application.

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h. What is the CLI command to find the outbound IP addresses of a web app?

Ans: `az webapp show \`
 `--resource-group <group_name> \`
 `--name <app_name> \`
 `--query outboundIpAddresses \`

i. The runtime that your application has been build is not available in azure app service. You must deploy your app in app service. How would you remediate this situation?

Ans: Deploy as a container and include the runtime in the image